B.Tech 2nd Year

IT Workshop-I

Lab 1 — Introduction to Java: Basics, Operators, Control Flow & Loops

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General Instructions

1. Write, compile, and run each program in a separate Java file. The file name must match the public class name exactly (e.g., Calculator.java).
2. Take input from the user by Scanner.
3. Show the output of each program (as a comment or a screenshot, as instructed by TAs) after the code.
4. Use meaningful variable names and add comments explaining each step.
5. Attempt all questions in the order given.
6. Explore the official GitHub repo shared in the Google Classroom for installation and quick usage document

Questions

Q1. Write a Java program that declares two hardcoded integer values and performs addition, subtraction, multiplication, division, and modulus operations on them. Display all five results with appropriate labels.

- Example values: a = 15, b = 4
- Expected outputs: Sum, Difference, Product, Quotient, and Remainder

Q2. Write a Java program that calculates and displays the area and perimeter of a rectangle taking input from users values for length and breadth.

- Example values: length = 12.5, breadth = 6.0
- Formulas: Area = length × breadth, Perimeter = 2 × (length + breadth)

Q3. Write a Java program that calculates and displays the area and circumference of a circle .

- Example value: radius = 7.0
- Formulas: Area = πr^2, Circumference = $2\pi r$ (use $\pi \approx 3.14159$)

Q4. Write a Java program that checks whether a number is a prime number or not, and displays an appropriate message. Use a loop and a conditional statement in your logic.

- Example value: number = 29
- Hint: A number is prime if it is greater than 1 and has no divisors other than 1 and itself.

Q5. Write a Java program that checks whether a number is even or odd using an if-else statement, and displays an appropriate message.

- Example value: number = 18
- Hint: Use the modulus operator (%) to test divisibility by 2.

Q6. Write a Java program that finds and displays the largest among three numbers using if-else statements.

- Example values: a = 23, b = 45, c = 10
- Hint: Use nested if-else or else-if statements to compare the values pairwise.

Q7. Write a Java program that checks whether a year is a leap year or not, using if-else statements with logical operators, and displays an appropriate message.

- Example value: year = 2024
- Hint: A year is a leap year if it is divisible by 4 and (not divisible by 100 or divisible by 400).

Q8. Write a Java program that calculates and displays the simple interest and total amount for a principal, rate of interest, and time period.

- Example values: principal = 10000.0, rate = 7.5, time = 3
- Formulas: Simple Interest = (P × R × T) / 100, Total Amount = Principal + Simple Interest

Q9. Write a Java program that swaps two numbers without using a temporary (third) variable, and displays the values before and after swapping.

- Example values: a = 5, b = 9
- Hint: Use arithmetic (addition/subtraction) or the XOR bitwise operator to swap the values.

Q10. Write a Java program that displays the name of the day of the week corresponding to a day number (1 to 7), using a switch-case statement.

- Example value: day = 4
- Hint: Use cases 1 through 7 for Monday through Sunday, and a default case to handle invalid values.

Q11. Write a Java program for converting binary to decimal and vice versa take input from the user.

- Sample input (for testing): binary number = 1011, decimal number = 25
- Hint: Use Integer.parseInt(str, 2) to convert binary to decimal, and Integer.toBinaryString(num) to convert decimal to binary.